

|          |                                                                                      |           |
|----------|--------------------------------------------------------------------------------------|-----------|
|          |                                                                                      |           |
| 2(a)     | $F = 1050 \times 9.81 \times 0.21V$ <b>or</b> $W = \rho \times 9.81 \times V$        | <b>C1</b> |
|          | $0.21V \times 1050 (\times 9.81) = V (\times 9.81) \times \rho$                      | <b>A1</b> |
|          | $\rho = 220 \text{ kg m}^{-3}$                                                       |           |
| 2(b)(i)  | $F = 1050 \times 9.81 \times V$ <b>or</b> $W = 220 \times 9.81 \times V$             | <b>C1</b> |
|          | $(V \times 1050 \times 9.81) - (V \times 220 \times 9.81) = (V \times 220) \times a$ | <b>C1</b> |
|          | $a = 37 \text{ m s}^{-2}$                                                            | <b>A1</b> |
| 2(b)(ii) | the (downward) drag / viscous force increases (with speed)                           | <b>M1</b> |
|          | resultant force decreases (as upthrust and weight remain the same)                   | <b>M1</b> |
|          | acceleration decreases (as its velocity increases)                                   | <b>A1</b> |